

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks Available |     |     |       |       |      |
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|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 3        | (a) |  |  | Any 2 × (1) from:<br>Green plants {use chlorophyll / have chloroplasts} (1)<br>to absorb light energy (1)<br>and convert carbon dioxide and water (1)<br>into <u>glucose</u> and oxygen (1)<br>AND<br>Photosynthesis can't occur / leaves are black / no chlorophyll (1)<br>Ash trees unable to make {food/glucose} (so they are unable to grow/die) (1)                                                                                                                                                                                                                                                                                                  | 3               | 1   |     | 4     |       |      |
|          | (b) |  |  | <b>Indicative content</b> <ul style="list-style-type: none"><li>• start with lamp near/far away from <i>Elodea</i></li><li>• count number of bubbles in a certain time</li><li>• change position of lamp</li><li>• count bubbles again</li><li>• keep temperature and sodium hydrogen carbonate conc. constant</li><li>• Repeat the experiment at different temperatures</li><li>• with the lamp in a constant position</li><li>• Keep sodium hydrogen carbonate conc. constant</li><li>• Repeat the experiment at different sodium hydrogen carbonate conc.</li><li>• with the lamp in a constant position</li><li>• with temperature constant</li></ul> |                 |     | 6   | 6     |       | 6    |

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks Available |     |     |       |       |      |
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|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  |  |  | <p><b>5-6 marks</b><br/>A full description for at least two limiting factors<br/><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> <p><b>3-4 marks</b><br/>A full description for one limiting factor including one or two controlled variables<br/>Or<br/>Part descriptions for two or more limiting factors which may not included any controlled variables<br/><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i></p> <p><b>1-2 marks</b><br/>Part of method described for one limiting factor<br/>Or<br/>List of limiting factors given <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i></p> <p><b>0 marks</b><br/><i>No attempt made or no response worthy of credit.</i></p> |                 |     |     |       |       |      |
|          |  |  |  | Question 3 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3               | 1   | 6   | 10    | 0     | 6    |